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Bayesian model specifications

For $\mathcal{N}(m, v)$, the second argument denotes variance. Half-normal distributions are specified by their scale parameters.

Unconstrained normal population model

$$\begin{aligned}
 y_{ij} \mid \theta_i &\sim \mathcal{N}(\theta_i, \sigma_\epsilon^2), \\
 \theta_i &\sim \mathcal{N}(\mu_\theta, \sigma_\theta^2), \\
 \log \mu_\theta &\sim \mathcal{N}(0.5, 1^2), \\
 z &\sim \mathcal{N}\left(-\frac{2}{3}, \left(\frac{2}{3}\right)^2\right), \\
 p &= \frac{\Phi(z)}{2}, \\
 \sigma_\theta &= -\frac{\mu_\theta}{\Phi^{-1}(p)}, \\
 \pi(\sigma_\epsilon) &\propto \sigma_\epsilon^{-2}, \quad \sigma_\epsilon > 0.
 \end{aligned}$$

Approximate dominance model

The specification was identical to the unconstrained model except that:

$$z \sim \mathcal{N}(-3.5, 0.5^2).$$

This concentrated prior probability on small negative-effect proportions.

Strictly positive log-normal population model

$$\begin{aligned}y_{ij} \mid \theta_i &\sim \mathcal{N}(\theta_i, \sigma_\epsilon^2), \\ \log \theta_i &\sim \mathcal{N}(\mu_{\log}, \sigma_{\log}^2), \\ \mu_{\log} &\sim \mathcal{N}(0.5, 1^2), \\ \sigma_{\log} &\sim \text{HalfNormal}(1), \\ \sigma_\epsilon &\sim \text{HalfNormal}(5), \\ P(\theta_i < 0) &= 0.\end{aligned}$$

Population summaries on the original spatial scale were derived as:

$$\begin{aligned}\mu_\theta &= \exp\left(\mu_{\log} + \frac{\sigma_{\log}^2}{2}\right), \\ \sigma_\theta &= \sqrt{[\exp(\sigma_{\log}^2) - 1] \exp(2\mu_{\log} + \sigma_{\log}^2)}.\end{aligned}$$